Exp 12: Implementing Security Features and Governance / Administration Practices in Zoho Creator

Register Number: 241401127 | Student Name: Vijay Kumar C K

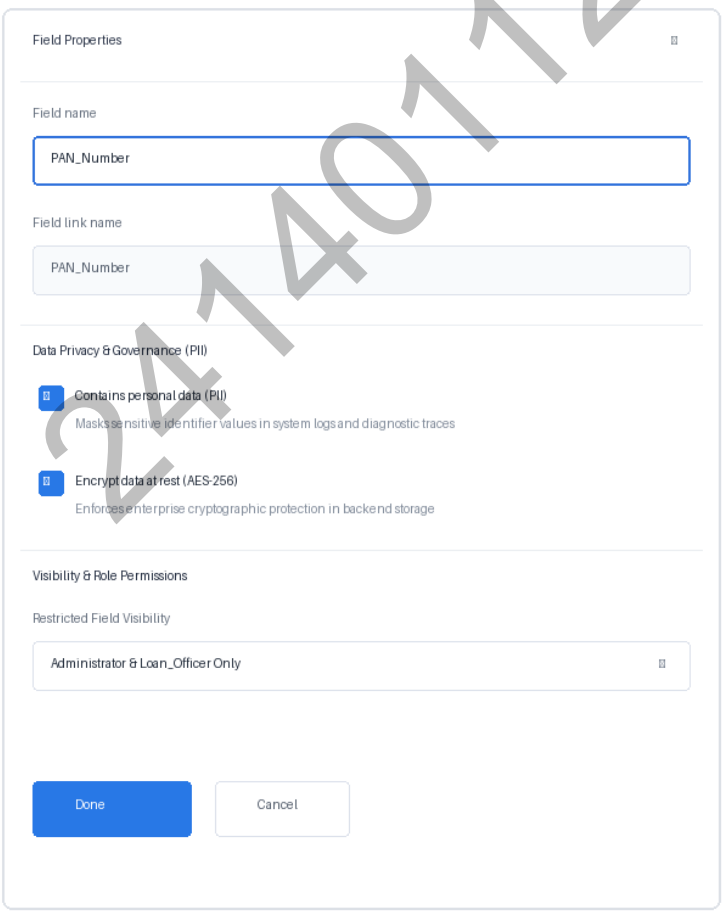
1. Aim

To implement, configure, and evaluate enterprise security features, Personal Identifiable Information (PII) data privacy protection, audit logging trails, IP restrictions, and administrative governance policies within the Loan Application & Pre-Screening Management Zoho Creator application.

2. Procedure

Step 1: PII Data Privacy & Field Encryption Configuration

Navigated to Form Builder for Loan_Application_Form and identified sensitive customer financial and identity attributes (Applicant_Name, PAN_Number, Email_Address, Contact_Number, and Credit_Score). Accessing Field Properties, configured the Personal Data (PII) flag to enforce encryption at rest (AES-256), mask plain-text output in system logs, and comply with institutional data privacy regulations (e.g., RBI cybersecurity guidelines and GDPR).



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Figure 1: Field Properties panel for PAN_Number showing PII flag and AES-256 encryption at rest enabled

Step 2: Enable System Audit Trail & Activity Logging

Navigated to Zoho Creator Setup > Admin Governance > Audit Logs. Enabled comprehensive tracking for user logins, record lifecycle events (Create, Update, Delete), export requests, and workflow execution triggers to establish full operational accountability, traceability, and non-repudiation.



Zoho Creator Setup > Admin Governance > Security Audit Trail — Application: Loan Management

Filter by Module: [Loan_Application_Form] Timeframe: [Past 30 Days] Status: [All]

Timestamp	User Account	Action Type	Affected Entity / Event	IP Address	Status
2026-10-18 10:14:22	admin@loanportal.internal	Schema Modified	Added Internal_Notes (Admin Only)	192.168.1.104	SUCCESS
2026-10-18 11:32:05	loan_officer_1@bank.internal	Record Viewed	Loan_Application_Form (ID: 15032)	192.168.1.112	SUCCESS
2026-10-18 14:05:41	admin@loanportal.internal	Stage Deployment	Promoted Version 1.0 -> 2.0	192.168.1.104	SUCCESS
2026-10-18 16:20:19	applicant_auth_svc	Record Inserted	Loan_Application_Form (Ram Kumar)	Internal Gateway	SUCCESS
2026-10-19 09:44:11	admin@loanportal.internal	Export Blocked	Unmask PAN & Credit_Score Request	192.168.1.104	DENIED (PII)
2026-10-19 12:10:04	system_worker	Workflow Executed	Loan_Application_Income_alert Trigger	Cloud Scheduler	SUCCESS

Figure 2: Zoho Creator Admin Governance audit log console demonstrating activity tracking for Loan_Application_Form

Step 3: Configure IP Restrictions & Authentication Policies

In Security & Governance Settings, established corporate network IP allowlisting policies (Subnet 192.168.1.0/24) to restrict backend administrative and loan officer console access. Mandated Multi-Factor Authentication (MFA) enforcement across all administrator and officer roles to prevent unauthorized access.

Step 4: Establish Backup Schedules & Data Archival Governance

Configured scheduled automated weekly backups with encrypted snapshot storage, and defined formal data retention rules for inactive historical loan application logs to satisfy regulatory audit requirements.

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3. Output

3.1 Sensitive Data (PII) Protection & Encryption Matrix

Field-level security and compliance configuration applied to Loan Application attributes:

Field Name	Classification	Security Control Applied	Storage State	Log Visibility
Applicant_Name	PII / Identity	Data Privacy Flag Enabled	Encrypted at Rest	Masked in System Traces
PAN_Number	PII / Financial ID	AES-256 Field Encryption	Encrypted at Rest	Partially Masked (******127A)

Contact_Number	PII / Contact	Data Privacy Flag Enabled	Encrypted at Rest	Partially Masked (*******432)
Email_Address	PII / Contact	AES-256 Field Encryption	Encrypted at Rest	Restricted to Authorized Roles
Credit_Score	Sensitive Financial	Access-Restricted Field	Encrypted at Rest	Hidden from Public Reports

3.2 System Audit Trail & Administrative Event Logging

Sample audit trail log demonstrating tracking of administrative actions and record modifications on the loan application:

Timestamp	User Account	Action Type	Affected Entity / Event	IP Address	Status
2026-10-18 10:14:22	admin@loanportal.internal	Schema Modified	Added Internal_Notes (Admin Only)	192.168.1.104	SUCCESS
2026-10-18 11:32:05	loan_officer_1@bank.internal	Record Viewed	Loan_Application_Form (ID: 15032)	192.168.1.112	SUCCESS
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2026-10-19 09:44:11	admin@loanportal.internal	Export Blocked	Unmask PAN & Credit_Score Request	192.168.1.104	DENIED (PII)

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3.3 Enterprise Security & Administrative Governance Policies

Summary of governance controls and compliance policies active across the application environment:

Governance Domain	Policy Standard	Configuration State	Compliance Objective
Authentication	Multi-Factor Authentication (MFA)	Enforced for Admins & Officers	Prevents unauthorized credential takeover
Network Access	IP Whitelisting Range	Subnet 192.168.1.0/24	Restricts administrative console access to corporate subnet
Session Management	Idle Session Timeout	15 Minutes	Mitigates session hijacking and unattended terminal

			risks
Data Archival	Automated Scheduled Backup	Weekly Full Encrypted Snapshot	Ensures business continuity & disaster recovery readiness

4. Result / Observations

Security and governance features were successfully configured and verified in Zoho Creator for the Loan Application & Pre-Screening Management application:

1. Sensitive customer attributes (PAN, Phone, Email, Credit Score) were secured with field-level AES-256 encryption and PII data protection flags.
2. System audit logs recorded administrative configurations and record lifecycles, ensuring operational traceability and compliance auditing.
3. Authentication and IP perimeter controls established robust governance boundaries protecting enterprise banking data.

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